

Statistical Inference

Series 2 • Complete Solutions

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Note: This document contains my solutions to every exercise in the Series 2 problem set. If you are lost and are looking for solutions to only the graded exercises, [click here](#).

Exercise 1

Suppose that we have a random sample $X_1, \dots, X_n$ i.i.d. $\sim f_\theta$. We write X (or $\tilde{X}_n$ to emphasize on the sample size) to denote the entire sample $(X_1, \dots, \tilde{X}_n)$. For the following scenarios, find the joint distribution $p_\theta(\tilde{x})$ (pdf/pmf) of $\tilde{X}$. Identify sufficient statistics for the unknown parameter θ in each case.

- (a) $f_\theta = \text{Bernoulli}(\theta)$, $\theta \in [0, 1]$.
- (b) $f_\theta = \text{Poisson}(\theta)$, $\theta > 0$.
- (c) $f_\theta = \text{Geometric}(\theta)$, $\theta \in [0, 1]$.
- (d) $f_\theta = \text{Uniform}(0, \theta)$, $\theta > 0$.
- (e) $f_\theta = \text{Uniform}(\theta, 1)$, $\theta < 1$.
- (f) $f_\theta = \text{Uniform}(\theta, \theta + 1)$, $\theta \in \mathbb{R}$.
- (g) $f_\theta = \text{Normal}(0, \theta^2)$, $\theta > 0$.
- (h) $f_\theta = \text{Normal}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$, $\theta = (\mu, \sigma^2)$.
- (i) $f_\theta = \text{Cauchy}(\theta, 1)$, $\theta \in \mathbb{R}$.
- (j) $f_\theta = \text{Cauchy}(0, \theta)$, $\theta > 0$.
- (k) $f_\theta = \text{Cauchy}(\mu, \sigma)$, $\mu \in \mathbb{R}$, $\sigma > 0$, $\theta = (\mu, \sigma)$.
- (l) $f_\theta = \text{Laplace}(\theta, 1)$, $\theta \in \mathbb{R}$.
- (m) $f_\theta = \text{Laplace}(0, \theta)$, $\theta > 0$.
- (n) $f_\theta = \text{Laplace}(\mu, \sigma)$, $\mu \in \mathbb{R}$, $\sigma > 0$, $\theta = (\mu, \sigma)$.
- (o) $f_\theta = \text{Normal}(\theta, \theta^2)$, $\theta \in \mathbb{R}$.
- (p) $f_\theta = \text{Normal}(\theta, \theta)$, $\theta > 0$.

Solution:

In every part below we write $x_{(1)} \leq \dots \leq x_{(n)}$ for the order statistics of the sample, and we

lean on the *Neyman-Fisher Factorization Theorem* throughout: if $p_\theta(x) = g_\theta(T(x)) h(x)$ for some functions g_θ and h not depending on θ , then $T(X)$ is sufficient for θ .

- (a) **Bernoulli**(θ). The pdf of a single observation is $f_\theta(x) = \theta^x(1 - \theta)^{1-x} \mathbb{1}_{\{x \in \{0,1\}\}}$. Since $X_1, \dots, X_n$ are i.i.d.,

$$p_\theta(x) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}$$

This is already of the form $g_\theta(T(x)) \times h(x)$ with $h(x) = 1$. Hence

$$T(x) = \sum_{i=1}^n x_i$$

is sufficient for θ .

- (b) **Poisson**(θ). Here $f_\theta(x) = e^{-\theta} \theta^x / x!$, so

$$p_\theta(x) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!} = \underbrace{e^{-n\theta} \theta^{\sum_{i=1}^n x_i}}_{g_\theta(T(x))} \times \underbrace{\frac{1}{\prod_{i=1}^n x_i!}}_{h(x)}$$

Hence

$$T(x) = \sum_{i=1}^n x_i$$

is sufficient for θ .

- (c) **Geometric**(θ). Taking X_i to be the number of trials until the first success, $f_\theta(x) = \theta(1 - \theta)^{x-1}$ for $x = 1, 2, \dots$. Then

$$p_\theta(x) = \prod_{i=1}^n \theta(1 - \theta)^{x_i-1} = \theta^n (1 - \theta)^{\sum_{i=1}^n x_i - n}$$

with $h(x) = 1$. Hence

$$T(x) = \sum_{i=1}^n x_i$$

is sufficient for θ .

- (d) **Uniform**($0, \theta$). Here $f_\theta(x) = \frac{1}{\theta} \mathbb{1}_{\{0 < x < \theta\}}$, so

$$p_\theta(x) = \prod_{i=1}^n \frac{1}{\theta} \mathbb{1}_{\{0 < x_i < \theta\}} = \frac{1}{\theta^n} \mathbb{1}_{\{0 < x_{(1)}\}} \mathbb{1}_{\{x_{(n)} < \theta\}}$$

since all n observations lie in $(0, \theta)$ if and only if the smallest one exceeds 0 and the largest one is below θ . Taking $g_\theta(T(x)) = \theta^{-n} \mathbb{1}_{\{x_{(n)} < \theta\}}$ and $h(x) = \mathbb{1}_{\{0 < x_{(1)}\}}$,

$$T(x) = x_{(n)} = \max_i x_i$$

is sufficient for θ .

(e) **Uniform**($\theta, 1$). Here $f_\theta(x) = \frac{1}{1-\theta} \mathbb{1}_{\{\theta < x < 1\}}$, so

$$p_\theta(\underset{\sim}{x}) = \frac{1}{(1-\theta)^n} \mathbb{1}_{\{\theta < x_{(1)}\}} \mathbb{1}_{\{x_{(n)} < 1\}}$$

Taking $g_\theta(T(\underset{\sim}{x})) = (1-\theta)^{-n} \mathbb{1}_{\{\theta < x_{(1)}\}}$ and $h(\underset{\sim}{x}) = \mathbb{1}_{\{x_{(n)} < 1\}}$,

$$\boxed{T(\underset{\sim}{x}) = x_{(1)} = \min_i x_i}$$

is sufficient for θ .

(f) **Uniform**($\theta, \theta + 1$). Here $f_\theta(x) = \mathbb{1}_{\{\theta < x < \theta + 1\}}$, so

$$\begin{aligned} p_\theta(\underset{\sim}{x}) &= \prod_{i=1}^n \mathbb{1}_{\{\theta < x_i < \theta + 1\}} = \mathbb{1}_{\{\theta < x_{(1)}\}} \mathbb{1}_{\{x_{(n)} < \theta + 1\}} \\ &= \mathbb{1}_{\{x_{(n)} - 1 < \theta < x_{(1)}\}} \end{aligned}$$

with $h(\underset{\sim}{x}) = 1$. Hence

$$\boxed{T(\underset{\sim}{x}) = (x_{(1)}, x_{(n)})}$$

is jointly sufficient for θ .

(g) **Normal**($0, \theta^2$). Here $f_\theta(x) = (2\pi\theta^2)^{-1/2} \exp\left(-\frac{x^2}{2\theta^2}\right)$, so

$$p_\theta(\underset{\sim}{x}) = (2\pi\theta^2)^{-n/2} \exp\left(-\frac{1}{2\theta^2} \sum_{i=1}^n x_i^2\right)$$

with $h(\underset{\sim}{x}) = 1$. Hence

$$\boxed{T(\underset{\sim}{x}) = \sum_{i=1}^n x_i^2}$$

is sufficient for θ .

(h) **Normal**(μ, σ^2). Here $f_\theta(x) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$, so

$$p_\theta(\underset{\sim}{x}) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right)$$

Expanding the square, $\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2$, so

$$p_\theta(\underset{\sim}{x}) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{n\mu^2}{2\sigma^2}\right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 + \frac{\mu}{\sigma^2} \sum_{i=1}^n x_i\right)$$

with $h(\underset{\sim}{x}) = 1$, and the exponent depends on $\underset{\sim}{x}$ only through $\sum x_i$ and $\sum x_i^2$. Hence

$$\boxed{T(\underset{\sim}{x}) = \left(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2\right)}$$

is jointly sufficient for $\theta = (\mu, \sigma^2)$.

- (i) **Cauchy**($\theta, 1$). Here $f_\theta(x) = \frac{1}{\pi(1 + (x - \theta)^2)}$, so

$$p_\theta(\underset{\sim}{x}) = \frac{1}{\pi^n} \prod_{i=1}^n \frac{1}{1 + (x_i - \theta)^2}$$

Unlike the previous parts, this product does not simplify to a function of any lower-dimensional summary of $\underset{\sim}{x}$ – for generic n, θ , no algebraic reduction of $\sum \log(1 + (x_i - \theta)^2)$ is possible. The one genuine simplification available is that $p_\theta(\underset{\sim}{x})$ is a *symmetric* function of $x_1, \dots, x_n$ (permuting the observations leaves it unchanged), so it depends on $\underset{\sim}{x}$ only through the order statistics, with $h(\underset{\sim}{x}) = 1$. Hence

$$\boxed{T(\underset{\sim}{x}) = (x_{(1)}, \dots, x_{(n)})}$$

(the full vector of order statistics) is sufficient for θ .

- (j) **Cauchy**($0, \theta$). Here $f_\theta(x) = \frac{\theta}{\pi(\theta^2 + x^2)}$, so

$$p_\theta(\underset{\sim}{x}) = \frac{\theta^n}{\pi^n} \prod_{i=1}^n \frac{1}{\theta^2 + x_i^2}$$

Here f_θ depends on x_i only through x_i^2 , and the product is symmetric in the observations. So $p_\theta(\underset{\sim}{x})$ depends on $\underset{\sim}{x}$ only through the (unordered) values $x_1^2, \dots, x_n^2$, equivalently the order statistics of $|X_1|, \dots, |X_n|$, with $h(\underset{\sim}{x}) = 1$. Hence

$$\boxed{T(\underset{\sim}{x}) = (|x|_{(1)}, \dots, |x|_{(n)})}$$

is sufficient for θ .

- (k) **Cauchy**(μ, σ). Here $f_\theta(x) = \frac{\sigma}{\pi(\sigma^2 + (x - \mu)^2)}$, so

$$p_\theta(\underset{\sim}{x}) = \frac{\sigma^n}{\pi^n} \prod_{i=1}^n \frac{1}{\sigma^2 + (x_i - \mu)^2}$$

As in part (i), no algebraic reduction is available beyond the symmetry of the product in $x_1, \dots, x_n$. With $h(\underset{\sim}{x}) = 1$,

$$\boxed{T(\underset{\sim}{x}) = (x_{(1)}, \dots, x_{(n)})}$$

is sufficient for $\theta = (\mu, \sigma)$.

- (l) **Laplace**($\theta, 1$). Here $f_\theta(x) = \frac{1}{2} \exp(-|x - \theta|)$, so

$$p_\theta(\underset{\sim}{x}) = \frac{1}{2^n} \exp\left(-\sum_{i=1}^n |x_i - \theta|\right)$$

The exponent $\sum_i |x_i - \theta|$ does not reduce to a function of a lower-dimensional statistic (which of the x_i lie above or below θ changes as θ varies). Since the expression is symmetric in $x_1, \dots, x_n$, with $h(\tilde{x}) = 1$,

$$\boxed{T(\tilde{x}) = (x_{(1)}, \dots, x_{(n)})}$$

is sufficient for θ .

(m) **Laplace**(0, θ). Here $f_\theta(x) = \frac{1}{2\theta} \exp(-|x|/\theta)$, so

$$p_\theta(\tilde{x}) = \frac{1}{(2\theta)^n} \exp\left(-\frac{1}{\theta} \sum_{i=1}^n |x_i|\right)$$

with $h(\tilde{x}) = 1$. Hence

$$\boxed{T(\tilde{x}) = \sum_{i=1}^n |x_i|}$$

is sufficient for θ .

(n) **Laplace**(μ, σ). Here $f_\theta(x) = \frac{1}{2\sigma} \exp(-|x - \mu|/\sigma)$, so

$$p_\theta(\tilde{x}) = \frac{1}{(2\sigma)^n} \exp\left(-\frac{1}{\sigma} \sum_{i=1}^n |x_i - \mu|\right)$$

As in part (1), $\sum_i |x_i - \mu|$ does not admit a lower-dimensional reduction. With $h(\tilde{x}) = 1$,

$$\boxed{T(\tilde{x}) = (x_{(1)}, \dots, x_{(n)})}$$

is sufficient for $\theta = (\mu, \sigma)$.

(o) **Normal**(θ, θ^2). Here $f_\theta(x) = (2\pi\theta^2)^{-1/2} \exp\left(-\frac{(x - \theta)^2}{2\theta^2}\right)$, so

$$p_\theta(\tilde{x}) = (2\pi\theta^2)^{-n/2} \exp\left(-\frac{1}{2\theta^2} \sum_{i=1}^n (x_i - \theta)^2\right)$$

Expanding, $\sum_i (x_i - \theta)^2 = \sum_i x_i^2 - 2\theta \sum_i x_i + n\theta^2$, so the exponent involves $\tilde{x}$ only through $\sum x_i$ and $\sum x_i^2$. With $h(\tilde{x}) = 1$,

$$\boxed{T(\tilde{x}) = \left(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2\right)}$$

is jointly sufficient for θ .

(p) **Normal**(θ, θ). Here $f_\theta(x) = (2\pi\theta)^{-1/2} \exp\left(-\frac{(x-\theta)^2}{2\theta}\right)$, so

$$p_\theta(\tilde{x}) = (2\pi\theta)^{-n/2} \exp\left(-\frac{1}{2\theta} \sum_{i=1}^n (x_i - \theta)^2\right)$$

Expanding as before, the exponent again involves $\tilde{x}$ only through $\sum x_i$ and $\sum x_i^2$. With $h(\tilde{x}) = 1$,

$$T(\tilde{x}) = \left(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2 \right)$$

is jointly sufficient for θ . (We revisit this in Exercise 5 – it turns out this can be reduced further.)

Exercise 2

Consider three independent random variables $X_1 \sim \text{Uniform}(\theta - 1, \theta)$, $X_2 \sim \text{Uniform}(\theta, \theta + 1)$ and $X_3 \sim \text{Uniform}(\theta + 1, \theta + 2)$. Find a bivariate sufficient statistic for θ based on X_1, X_2, X_3 .

Solution:

We have the following pdf for each X_i

$$f_{X_1}(x_1) = \mathbb{1}_{\{\theta-1 < x_1 < \theta\}} \quad f_{X_2}(x_2) = \mathbb{1}_{\{\theta < x_2 < \theta+1\}} \quad f_{X_3}(x_3) = \mathbb{1}_{\{\theta+1 < x_3 < \theta+2\}}$$

Since X_i 's are independently distributed, we have the following joint pdf

$$\begin{aligned} f_{\tilde{X}}(\tilde{x}) &= \mathbb{1}_{\{\theta-1 < x_1 < \theta\}} \mathbb{1}_{\{\theta < x_2 < \theta+1\}} \mathbb{1}_{\{\theta+1 < x_3 < \theta+2\}} \\ &= \mathbb{1}_{\{x_1 < \theta < x_1+1\}} \mathbb{1}_{\{x_2-1 < \theta < x_2\}} \mathbb{1}_{\{x_3-2 < \theta < x_3-1\}} \\ &= (\mathbb{1}_{\{\theta > x_1\}} \mathbb{1}_{\{\theta > x_2-1\}} \mathbb{1}_{\{\theta > x_3-2\}}) \times (\mathbb{1}_{\{\theta < x_1+1\}} \mathbb{1}_{\{\theta < x_2\}} \mathbb{1}_{\{\theta < x_3-1\}}) \\ &= (\mathbb{1}_{\{\theta > \max\{x_1, x_2-1, x_3-2\}\}}) \times (\mathbb{1}_{\{\theta < \min\{x_1+1, x_2, x_3-1\}\}}) \end{aligned}$$

Now, we'll apply the *Neyman-Fisher Factorization Theorem* to the joint pdf. Thus, we have the following

$$f_{\tilde{X}}(\tilde{x}) = \underbrace{(\mathbb{1}_{\{\theta > \max\{x_1, x_2-1, x_3-2\}\}})}_{g_\theta(T(\tilde{x}))} \times \underbrace{(\mathbb{1}_{\{\theta < \min\{x_1+1, x_2, x_3-1\}\}})}_{h(\tilde{x})} \times \underbrace{1}_{h(\tilde{x})}$$

Thus, we have the following

$$g_\theta(T(\tilde{x})) = (\mathbb{1}_{\{\theta > \max\{x_1, x_2-1, x_3-2\}\}}) (\mathbb{1}_{\{\theta < \min\{x_1+1, x_2, x_3-1\}\}}) \quad \text{and} \quad h(\tilde{x}) = 1$$

Hence, according to Neyman-Fisher Factorization Theorem,

$$T(\tilde{x}) = (\max\{x_1, x_2 - 1, x_3 - 2\}, \min\{x_1 + 1, x_2, x_3 - 1\})$$

is jointly sufficient for θ

Exercise 3

Let $X_1, \dots, X_n$ be independent random variables with X_i having pdf

$$f_{\theta,i}(x) = \begin{cases} \frac{1}{2i\theta} & \text{if } -i(\theta - 1) < x < i(\theta + 1), \\ 0 & \text{otherwise.} \end{cases}$$

Write down the joint pdf $p_{\theta}(\tilde{x})$ of $X_1, \dots, X_n$. Find a bivariate sufficient statistic for θ .

Solution:

We can rewrite the support condition for each X_i using an indicator. Noting that $-i(\theta - 1) = i(1 - \theta)$, we have

$$f_{\theta,i}(x_i) = \frac{1}{2i\theta} \mathbb{1}_{\{i(1-\theta) < x_i < i(\theta+1)\}}$$

Since the X_i 's are independently distributed, the joint pdf is the product of the marginals

$$\begin{aligned} p_{\theta}(\tilde{x}) &= \prod_{i=1}^n \frac{1}{2i\theta} \mathbb{1}_{\{i(1-\theta) < x_i < i(\theta+1)\}} \\ &= \left(\frac{1}{2\theta}\right)^n \left(\frac{1}{\prod_{i=1}^n i}\right) \prod_{i=1}^n \mathbb{1}_{\{i(1-\theta) < x_i < i(\theta+1)\}} \end{aligned}$$

Now we rewrite each indicator as a condition on θ rather than on x_i . From $i(1 - \theta) < x_i$ we get $\theta > 1 - \frac{x_i}{i}$, and from $x_i < i(\theta + 1)$ we get $\theta > \frac{x_i}{i} - 1$. So

$$\mathbb{1}_{\{i(1-\theta) < x_i < i(\theta+1)\}} = \mathbb{1}_{\{\theta > 1 - \frac{x_i}{i}\}} \mathbb{1}_{\{\theta > \frac{x_i}{i} - 1\}}$$

Taking the product over $i = 1, \dots, n$, and using $\prod_{i=1}^n i = n!$,

$$\begin{aligned} p_{\theta}(\tilde{x}) &= \left(\frac{1}{2\theta}\right)^n \frac{1}{n!} \prod_{i=1}^n \mathbb{1}_{\{\theta > 1 - \frac{x_i}{i}\}} \mathbb{1}_{\{\theta > \frac{x_i}{i} - 1\}} \\ &= \left(\frac{1}{2\theta}\right)^n \frac{1}{n!} \mathbb{1}_{\{\theta > \max_{1 \leq i \leq n} (1 - \frac{x_i}{i})\}} \mathbb{1}_{\{\theta > \max_{1 \leq i \leq n} (\frac{x_i}{i} - 1)\}} \end{aligned}$$

since a conjunction of conditions of the same type $\theta > c_i$ holds simultaneously if and only if θ exceeds the largest c_i .

Now we apply the *Neyman-Fisher Factorization Theorem*. We have

$$p_{\theta}(\tilde{x}) = \underbrace{\left(\frac{1}{2\theta}\right)^n \mathbb{1}_{\{\theta > \max_i (1 - \frac{x_i}{i})\}} \mathbb{1}_{\{\theta > \max_i (\frac{x_i}{i} - 1)\}}}_{g_{\theta}(T(\tilde{x}))} \times \underbrace{\frac{1}{n!}}_{h(\tilde{x})}$$

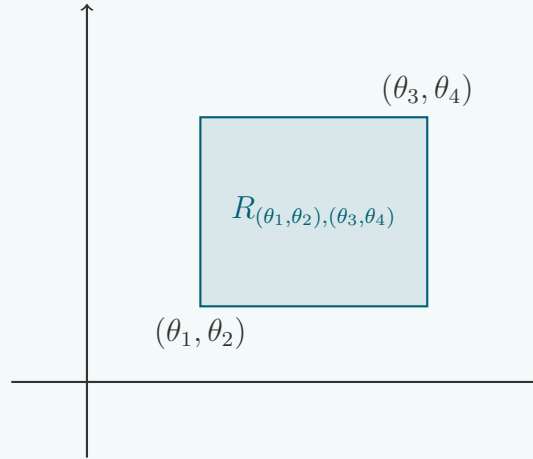
Hence, according to the Neyman-Fisher Factorization Theorem,

$$T(\tilde{x}) = \left(\max_{1 \leq i \leq n} \left(1 - \frac{x_i}{i}\right), \max_{1 \leq i \leq n} \left(\frac{x_i}{i} - 1\right) \right)$$

is jointly sufficient for θ

Exercise 4

Let $\theta_1, \theta_2, \theta_3, \theta_4$ be parameters satisfying $\theta_1 < \theta_3$ and $\theta_2 < \theta_4$. Denote by $R_{(\theta_1, \theta_2), (\theta_3, \theta_4)}$ the rectangular region with lower left corner (θ_1, θ_2) and upper right corner (θ_3, θ_4) (see the figure below).



For $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$, let $f_\theta(x, y)$ be the pdf of the uniform distribution on $R_{(\theta_1, \theta_2), (\theta_3, \theta_4)}$. That is

$$f_\theta(x, y) = \begin{cases} c & \text{if } (x, y) \in R_{(\theta_1, \theta_2), (\theta_3, \theta_4)}, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine the value of c .
- (b) Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. bivariate observations with density $f_\theta(x, y)$. Derive the joint pdf of the random sample $(X_1, Y_1), \dots, (X_n, Y_n)$.
- (c) Find a four-dimensional sufficient statistic for $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$ based on the random sample $(X_1, Y_1), \dots, (X_n, Y_n)$.

Solution:

- (a) First we'll calculate the area, S for the region. We can clearly see that the area is rectangular in nature with the lengths $\theta_3 - \theta_1$ and $\theta_4 - \theta_2$. Thus, the area of a rectangle with the following lengths is $area(S) = (\theta_3 - \theta_1)(\theta_4 - \theta_2)$. Hence, the *bivariate density* of a pair random variable, (X, Y) , distributed uniformly over the

area S would be

$$f_{\theta}(x, y) = \begin{cases} \frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} & \text{if } (x, y) \in S \\ 0 & \text{otherwise} \end{cases}$$

Thus, clearly

$$c = \frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)}$$

(b) Alternatively, the bivariate density can also be written as

$$f_{\theta}(x, y) = \frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} (\mathbb{1}_{\{\theta_1 < x < \theta_3\}} \mathbb{1}_{\{\theta_2 < y < \theta_4\}})$$

Since $(X_1, Y_1), \dots, (X_n, Y_n)$ are independently distributed bivariate, the joint density is just the product of their marginal densities. Hence, we have the following joint density for the given random sample

$$\begin{aligned} f_{\theta} \left(\underset{\sim}{x}, \underset{\sim}{y} \right) &= \prod_{i=1}^n \frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} (\mathbb{1}_{\{\theta_1 < x_i < \theta_3\}} \mathbb{1}_{\{\theta_2 < y_i < \theta_4\}}) \\ &= \left(\frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} \right)^n \prod_{i=1}^n (\mathbb{1}_{\{\theta_1 < x_i\}}) (\mathbb{1}_{\{\theta_2 < y_i\}}) (\mathbb{1}_{\{\theta_3 > x_i\}}) (\mathbb{1}_{\{\theta_4 > y_i\}}) \\ &= \left(\frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} \right)^n \mathbb{1}_{\{\theta_1 < x_{(1)}\}} \mathbb{1}_{\{\theta_2 < y_{(1)}\}} \mathbb{1}_{\{\theta_3 > x_{(n)}\}} \mathbb{1}_{\{\theta_4 > y_{(n)}\}} \end{aligned}$$

Thus, we have the following result for the joint pdf of n i.i.d bivariate uniformly distributed random variables (X_i, Y_i) in the given area S

$$f_{\theta} \left(\underset{\sim}{x}, \underset{\sim}{y} \right) = \left(\frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} \right)^n \mathbb{1}_{\{\theta_1 < x_{(1)}\}} \mathbb{1}_{\{\theta_2 < y_{(1)}\}} \mathbb{1}_{\{\theta_3 > x_{(n)}\}} \mathbb{1}_{\{\theta_4 > y_{(n)}\}}$$

where $x_{(i)}$ and $y_{(i)}$ are order statistic

(c) To find a sufficient statistic for $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$, we'll use the Neyman-Fisher Factorization Theorem. Consider the joint pdf in the following form

$$f_{\theta} \left(\underset{\sim}{x}, \underset{\sim}{y} \right) = \underbrace{\left[\left(\frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} \right)^n \mathbb{1}_{\{\theta_1 < x_{(1)}\}} \mathbb{1}_{\{\theta_2 < y_{(1)}\}} \mathbb{1}_{\{\theta_3 > x_{(n)}\}} \mathbb{1}_{\{\theta_4 > y_{(n)}\}} \right]}_{g_{\theta}(\underset{\sim}{x}, \underset{\sim}{y})} \times \underbrace{1}_{h(\underset{\sim}{x}, \underset{\sim}{y})}$$

Thus, according to Factorization Theorem, we can conclude that

$$T \left(\underset{\sim}{x}, \underset{\sim}{y} \right) = (x_{(1)}, x_{(n)}, y_{(1)}, y_{(n)})$$

are *jointly sufficient* for θ , where $x_{(i)}$ and $y_{(i)}$ are order statistic

Exercise 5

Recall that a sufficient statistic $T = T(X)$ for θ is said to be *minimal sufficient* for θ if, for any other sufficient statistic $S = S(X)$ of θ , we can write T as a function of S , i.e., there exists some function g such that $T(x) = g(S(x))$ for all $x \in \mathcal{X}$. We have also seen a condition which ensures that a statistic T is minimal sufficient. Consider Exercise 1. Verify whether the sufficient statistics that you obtained in that exercise are minimal sufficient. If not, then try to identify a minimal sufficient statistic.

Solution:

We use the *Lehmann-Scheffé likelihood ratio criterion*: T is minimal sufficient if and only if, for any two sample points $\tilde{x}$ and $\tilde{y}$, the ratio

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})}$$

is *free of θ* if and only if $T(\tilde{x}) = T(\tilde{y})$. We check each part of Exercise 1 against this criterion.

(a) **Bernoulli**(θ).

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})} = \frac{\theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}}{\theta^{\sum y_i} (1 - \theta)^{n - \sum y_i}} = \left(\frac{\theta}{1 - \theta} \right)^{\sum x_i - \sum y_i}$$

This is free of θ iff $\sum x_i = \sum y_i$. Hence $T(\tilde{x}) = \sum x_i$ is **minimal sufficient**.

(b) **Poisson**(θ).

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})} = \theta^{\sum x_i - \sum y_i} \times \frac{\prod y_i!}{\prod x_i!}$$

which is free of θ iff $\sum x_i = \sum y_i$. Hence $T(\tilde{x}) = \sum x_i$ is **minimal sufficient**.

(c) **Geometric**(θ).

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})} = (1 - \theta)^{\sum x_i - \sum y_i}$$

which is free of θ iff $\sum x_i = \sum y_i$. Hence $T(\tilde{x}) = \sum x_i$ is **minimal sufficient**.

(d) **Uniform**($0, \theta$). Assuming both samples have all positive entries,

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})} = \frac{\mathbb{1}_{\{x_{(n)} < \theta\}}}{\mathbb{1}_{\{y_{(n)} < \theta\}}}$$

If $x_{(n)} \neq y_{(n)}$, say $x_{(n)} < y_{(n)}$, then for θ between the two, this ratio is 0 at some θ and 1 at others, so it is not free of θ . Hence the ratio is free of θ iff $x_{(n)} = y_{(n)}$, so $T(\tilde{x}) = x_{(n)}$ is **minimal sufficient**.

- (e) **Uniform**($\theta, 1$). By the same argument as (d), applied to $x_{(1)}$ in place of $x_{(n)}$, $T(x) = x_{(1)}$ is **minimal sufficient**.

- (f) **Uniform**($\theta, \theta + 1$).

$$\frac{p_\theta(\tilde{x})}{p_\theta(\tilde{y})} = \frac{\mathbb{1}_{\{x_{(n)}-1 < \theta < x_{(1)}\}}}{\mathbb{1}_{\{y_{(n)}-1 < \theta < y_{(1)}\}}}$$

This is free of θ (equal to 1 for all θ where the denominator is defined) iff the two intervals $(x_{(n)} - 1, x_{(1)})$ and $(y_{(n)} - 1, y_{(1)})$ coincide, i.e., iff $x_{(1)} = y_{(1)}$ and $x_{(n)} = y_{(n)}$. Hence $T(x) = (x_{(1)}, x_{(n)})$ is **minimal sufficient**.

- (g) **Normal**($0, \theta^2$).

$$\frac{p_\theta(\tilde{x})}{p_\theta(\tilde{y})} = \exp\left(-\frac{1}{2\theta^2} \left(\sum x_i^2 - \sum y_i^2\right)\right)$$

which is free of θ iff $\sum x_i^2 = \sum y_i^2$. Hence $T(x) = \sum x_i^2$ is **minimal sufficient**.

- (h) **Normal**(μ, σ^2).

$$\frac{p_\theta(\tilde{x})}{p_\theta(\tilde{y})} = \exp\left(-\frac{1}{2\sigma^2} \left[\left(\sum x_i^2 - \sum y_i^2\right) - 2\mu\left(\sum x_i - \sum y_i\right)\right]\right)$$

Since $1/\sigma^2$ and μ/σ^2 range over linearly independent values as (μ, σ^2) varies, this is free of θ iff both $\sum x_i = \sum y_i$ and $\sum x_i^2 = \sum y_i^2$. Hence $T(x) = (\sum x_i, \sum x_i^2)$ is **minimal sufficient**.

- (i) **Cauchy**($\theta, 1$).

$$\frac{p_\theta(\tilde{x})}{p_\theta(\tilde{y})} = \prod_{i=1}^n \frac{1 + (y_i - \theta)^2}{1 + (x_i - \theta)^2}$$

This is a ratio of two degree- $2n$ polynomials in θ ; it is free of θ (constant) only if the two polynomials are proportional, which happens iff they have the same roots, i.e., iff $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_n\}$ agree as multisets, i.e., iff $x_{(i)} = y_{(i)}$ for all i . Hence $T(x) = (x_{(1)}, \dots, x_{(n)})$ is **minimal sufficient** – no further reduction is possible for the Cauchy location family.

- (j) **Cauchy**($0, \theta$).

$$\frac{p_\theta(\tilde{x})}{p_\theta(\tilde{y})} = \prod_{i=1}^n \frac{\theta^2 + y_i^2}{\theta^2 + x_i^2}$$

By the same polynomial-in- θ^2 argument as (i), this is free of θ iff $\{x_1^2, \dots, x_n^2\} = \{y_1^2, \dots, y_n^2\}$ as multisets, i.e., iff the order statistics of $|x_i|$ match those of $|y_i|$. Hence $T(x) = (|x|_{(1)}, \dots, |x|_{(n)})$ is **minimal sufficient**.

- (k) **Cauchy**(μ, σ). Identical reasoning to (i), now with two free parameters (μ, σ) giving at least as many constraints; the order statistics $T(x) = (x_{(1)}, \dots, x_{(n)})$ is **minimal sufficient**.

(l) **Laplace**($\theta, 1$).

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})} = \exp \left(\sum_{i=1}^n |y_i - \theta| - \sum_{i=1}^n |x_i - \theta| \right)$$

The function $\theta \mapsto \sum_i |x_i - \theta|$ is piecewise linear with a kink at every x_i ; two such functions (for $\tilde{x}$ and $\tilde{y}$) differ by a constant for all θ only if the underlying sets of kink locations – and hence the order statistics – coincide. Hence $T(\tilde{x}) = (x_{(1)}, \dots, x_{(n)})$ **is minimal sufficient**.

(m) **Laplace**($0, \theta$).

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})} = \exp \left(-\frac{1}{\theta} \left(\sum |x_i| - \sum |y_i| \right) \right)$$

which is free of θ iff $\sum |x_i| = \sum |y_i|$. Hence $T(\tilde{x}) = \sum |x_i|$ **is minimal sufficient**.

(n) **Laplace**(μ, σ). By the same kink-location argument as (l), now with an extra scale parameter σ , $T(\tilde{x}) = (x_{(1)}, \dots, x_{(n)})$ **is minimal sufficient**.

(o) **Normal**(θ, θ^2).

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})} = \exp \left(-\frac{1}{2\theta^2} \left(\sum x_i^2 - \sum y_i^2 \right) + \frac{1}{\theta} \left(\sum x_i - \sum y_i \right) \right)$$

Since $1/\theta^2$ and $1/\theta$ are linearly independent functions of θ , this is free of θ iff *both* coefficients vanish, i.e., iff $\sum x_i = \sum y_i$ and $\sum x_i^2 = \sum y_i^2$. Hence $T(\tilde{x}) = (\sum x_i, \sum x_i^2)$ **is minimal sufficient**.

(p) **Normal**(θ, θ).

$$\frac{p_{\theta}(\tilde{x})}{p_{\theta}(\tilde{y})} = \exp \left(-\frac{1}{2\theta} \left(\sum x_i^2 - \sum y_i^2 \right) + \left(\sum x_i - \sum y_i \right) \right)$$

Here only *one* power of θ appears, so this ratio is free of θ iff $\sum x_i^2 = \sum y_i^2$ *alone* – the leftover term $\exp(\sum x_i - \sum y_i)$ does not involve θ at all, so it places no constraint. This is a genuine difference from part (o): here the sufficient statistic found in Exercise 1, $T(\tilde{x}) = (\sum x_i, \sum x_i^2)$, is **not minimal**, since $\sum x_i$ can differ between $\tilde{x}$ and $\tilde{y}$ while the ratio still stays free of θ . The minimal sufficient statistic is just

$$\boxed{T(\tilde{x}) = \sum_{i=1}^n x_i^2}$$

The intuition: because the variance equals θ (not θ^2) here, θ enters the exponent through only a single power, so only one linear constraint on $(\tilde{x}, \tilde{y})$ survives – one dimension less than in part (o).